Project Description:-The system uses an Arduino Uno, potentiometers to simulate voltage and current inputs, a 16×2 LCD, a relay, LED indicators, and a buzzer. A state-machine-based firmware controls protection, fault indication, and reconnection.

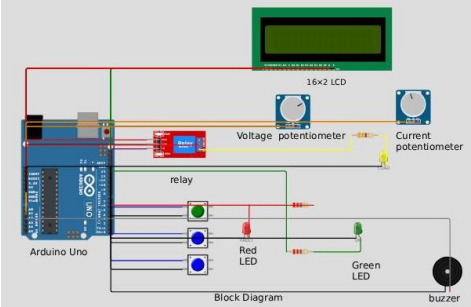


Fig:-Block-Diagram

Operating Conditions: The system is configured with an overvoltage threshold of 250 V, an undervoltage threshold of 180 V, and an overcurrent limit of 5 A. It detects overvoltage, undervoltage, overcurrent, and short-circuit conditions.

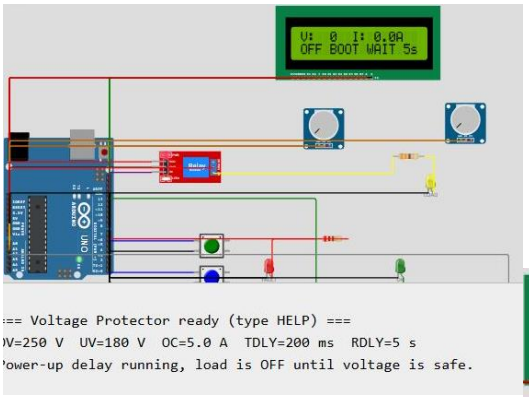






Fig:-Normal Condition Figure:-Under-voltage Figure:-Over-voltage

Results: The project was implemented and tested in Wokwi. Simulation tests demonstrated the system's response to abnormal input conditions by disconnecting the load and indicating faults.

Conclusion:-The project demonstrates a configurable embedded protection system and provides practical experience in Arduino programming, sensor-input simulation, relay control, and fault-handling logic.